Resolving Dual Processes in Complex Oscillatory Yielding

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The complex two-step yielding observed in some soft materials under oscillatory shearing is shown to result from two independent phenomena. The first step at small deformations corresponds to elastic softening, while the larger deformation feature corresponds to yielding. This interpretation is supported by experimental recovery rheology data and the construction of a rheo-physical model. Our findings elucidate the mechanisms governing complex yielding and underscore the insight afforded by recovery experiments, which decouple the underlying physics between recoverable and unrecoverable processes.

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The yielding transition in amorphous soft materials, where a response changes from that of a solid to that of a fluid under increasing load, is a topic of significant interest in both industrial applications and academic research and yet remains incompletely understood. While the microscopic underpinnings that lead to yielding differ for different materials, many of the macroscopic indicators remain consistent [1–6]. One important characteristic of the rheology of yield stress fluids that is seen across many different types of microstructures is a decrease in the storage modulus and an overshoot in the loss modulus as the amplitude increases [7–11]. Recently, an emerging paradigm called recovery rheology has been used to show that the loss modulus is composed of two separate terms, and that the overshoot in the loss modulus reflects a transition from recoverable to unrecoverable acquisition of strain [12]. Recovery rheology acknowledges that the total strain is a composite parameter that can be experimentally decomposed into recoverable and unrecoverable components, and has been used to understand complex phenomena [12–16], develop physics-based models that account for a wide range of yielding behaviors including a way to distinguish rates of yielding [17–19], and explain the physics underlying common empirical rheological rules [20,21].

While a decrease in the storage modulus and a smooth overshoot in the loss modulus have long been associated with yielding, some materials exhibit these features twice as the amplitude is increased. The appearance of these features has been interpreted as indicating that yielding takes place at two different length scales in separate steps and has been called "double yielding" or "two-step" yielding [6]. This two-step phenomenon has been observed in systems as diverse as emulsions [2,22–29], colloidal gels [3,30–40], glasses [4,41–49], suspensions [5,50–55], capillary gels [56,57], and magneto-responsive materials [58–60]. Despite the widespread reporting of the two-overshoot phenomenon over the past two decades, it remains incompletely understood experimentally, and models capable of predicting such behavior remain absent from the literature.

In this Letter, we demonstrate through experimental recovery rheology data and the construction of a rheo-physical model that the two overshoots result from distinct physical phenomena. We present data collected from oscillatory shear measurements of Carbopol dissolved in propylene glycol which exhibits attractive interactions. We show that the overshoot at larger deformation represents yielding and is associated with the acquisition of unrecoverable strain, consistent with recent studies of simple yield stress fluids [12,18]. In contrast, the overshoot at smaller deformations is caused by an elastic softening and is associated with changes in the way recoverable strain is acquired. These experimental observations are supported by the construction of a rheo-physical model that captures the double overshoot phenomenon and its frequency dependence.

The storage modulus and the loss modulus are proportional to the energy stored elastically and the energy dissipated viscously by a material. Additionally, the loss modulus is a composite parameter with solidlike and fluidlike contributions which relate to energy dissipated viscously by recoverable and unrecoverable processes, respectively. These terms are defined as

$$G'(\omega,\gamma_0)=\frac{2[\gamma_{\mathrm{rec}}(t)\sigma(t)]_{\mathrm{avg}}}{\gamma_0^2}, \tag{1}$$

$$G''(\omega,\gamma_0)=\frac{2[\dot{\gamma}_{\mathrm{tot}}(t)\sigma(t)]_{\mathrm{avg}}}{\omega\gamma_0^2}, \tag{2}$$

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$$G''_{\text{fluid}}(\omega,\gamma_0)=\frac{2[\dot{\gamma}_{\text{unrec}}(t)\sigma(t)]_{\text{avg}}}{\omega\gamma_0^2}=\frac{\sigma_0\gamma_s}{\gamma_0^2\cos(\psi)}, \quad (3)$$

$$G''_{\text{solid}}(\omega,\gamma_0)=\frac{2[\dot{\gamma}_{\text{rec}}(t)\sigma(t)]_{\text{avg}}}{\omega\gamma_0^2}=G''-G''_{\text{fluid}} \quad (4)$$

where γ_{rec}, $\dot{\gamma}_{\text{tot}}$, $\dot{\gamma}_{\text{unrec}}$, and $\dot{\gamma}_{\text{rec}}$ are the recoverable strain, the total strain rate, the unrecoverable strain rate, and recoverable strain rate, respectively, σ is the shear stress, γ_0 is the amplitude of the total strain, and ω is the angular frequency [12,61]. By analyzing the two-overshoot phenomenon in terms of the total loss modulus, previous works have been unable to distinguish energy dissipated by recoverable or unrecoverable processes. This distinction is important because it is the basis for why the effects in the present Letter differ from the understanding based on the total loss modulus.

While a strain-controlled measurement produces a stress that oscillates about zero, the stress-controlled experiments we have used produce strains that oscillate about a nonzero value. This value, called the strain shift and denoted γ_s, is directly associated with unrecoverable strain and has recently been shown to allow for calculation of the unrecoverable component of the loss modulus as seen in the second part of Eq. (3), where σ_0 is the amplitude of the applied stress, and ψ is the phase angle of the applied stress wave, which in this Letter is equal to zero [13,62]. The total and fluid components of the loss modulus allow us to calculate the solid component of the loss modulus, defined in Eq. (4).

We perform sinusoidal oscillatory shear tests to maximize the strain shift on a DHR 3 rheometer (TA Waters) using the arbitrary wave function with a 40 mm parallel plate geometry. Both plates have 220 grit sandpaper adhered to prevent wall slip. Samples of Carbopol Ultrez 10 (Lubrizol) are dissolved in propylene glycol (Dynalene), and mixed in an orbital mixer (THINKY) at 2000 RPM for 30 min, before resting 24 h prior to any experiments.

In a previous study, the strength of attractive interaction between constituent parts was controlled via the addition of salt to Carbopol in water [38]. In this Letter, we change the strength of attractive interaction by altering the polymer concentration. In Figs. 1(a) and 1(b), we show the total storage and loss modulus as a function of measured strain amplitude. The sample with a higher polymer concentration displayed a large storage and loss modulus at small amplitudes.

At the highest concentration of 2.0 wt%, we see only a single overshoot in the total loss modulus, while decreasing the concentration to 1.2 wt% leads to a response in which two overshoots can be clearly observed.

We measure the strain shift and use Eqs. (3) and (4) to calculate loss modulus components [62], and show the results in Figs. 1(c) and 1(d). Although the 2.0 wt% sample exhibits a single overshoot in the total loss modulus, as shown in Fig. 1(a), our methodology allows us to resolve two distinct contributions to this single peak, as shown in Fig. 1(c), where we observe overlapping peaks from the recoverable and unrecoverable contributions at intermediate amplitudes. By contrast, in the 1.2 wt% sample, shown in Fig. 1(b), the peaks in the recoverable and unrecoverable components are separated enough to be resolved in the total loss modulus, as seen in Fig. 1(d). In this case, changing the polymer concentration from 2.0 wt% to 1.2 wt% therefore only changes the underlying physics insofar as separating the two peaks enough to be resolvable in the total dynamic loss modulus. In addition to resolving the two peaks, our method also provides insight into their underlying physical causes. The overshoot at smaller amplitudes is reflective of an increase in energy dissipation due to a recoverable process, and is paired with a decrease in the storage modulus, whereas the second overshoot at larger amplitudes follows the previously reported transition to unrecoverable deformation typical of yield stress fluids [12]. We observe the same behavior across concentrations, which we show in the Supplemental Material [63]. Prior recovery rheology work has shown that the loss modulus can be decomposed experimentally into two terms and that yielding is associated with an overshoot in the unrecoverable part. We extend that understanding here to show that a single overshoot may also be the result of both parts of the loss modulus displaying overshoots, and that what has been called two-step yielding is a manifestations of the separation of those peaks so that their individual contributions can be resolved.

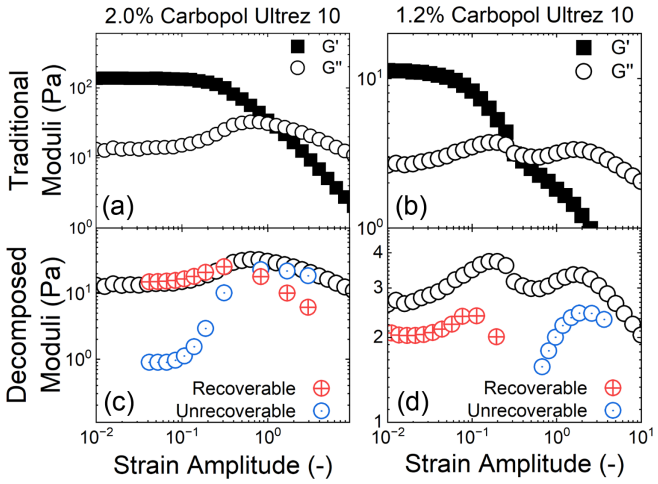


FIG. 1. Experimental data for Carbopol Ultrez 10 in propylene glycol at (a),(c) 2.0 wt% and (b),(d) 1.2 wt%. (a),(b) Show the total storage modulus (filled squares) and total loss modulus (open circles) as functions of the total strain amplitude, while (c), (d) show the total loss modulus (open circles), as well as the recoverable (red circles filled with plus sign) and unrecoverable (blue circles filled with dot) contributions. Vertical axes between samples for 2.0 wt% and 1.2 wt% are different, to contrast shapes more clearly.

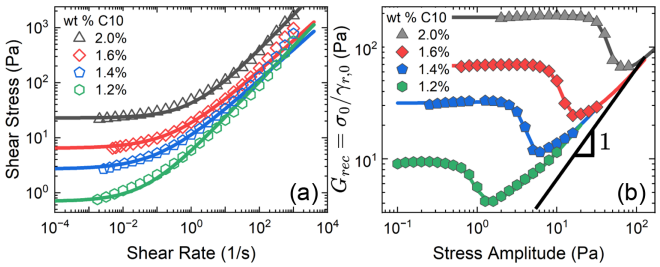


FIG. 2. (a) Stress as a function of shear rate at a steady state. Symbols represent experimental data for four concentrations of Carbopol Ultrez 10 in propylene glycol, while lines are fits to the Herschel-Bulkley model. (b) Recoverable modulus defined in Eq. (5) as a function of the applied shear stress. Symbols show experimental data, lines show data fittings following Eq. (6).

Having experimentally established that the two overshoots in the loss modulus and two decreases in the storage modulus are reflective of a sequence of behaviors consisting of softening followed by yielding, we now move onto the construction of a rheo-physical model capable of predicting such behavior. The construction of our model follows the same approach as our experimental methodology: we conceptually break the model response into recoverable and unrecoverable parts and then combine them to obtain the complete model. We begin our discussion of this construction with the unrecoverable behavior.

Within our model formulation for two-step yielding, the flow behavior and yield stress are clearly elucidated by the steady-shear flow curves, which we show in Fig. 2(a). Each datum is the long-time behavior measured at the end of transient steady shear startup tests across a range of shear rates. We allow at least 500% strain to be acquired at each shear rate to ensure that steady state is achieved. There are many ways to represent the behavior of yield stress fluids at steady state [64–66], and in the construction of our model, we choose to use the Herschel-Bulkley form, $\sigma = \sigma_y + k\dot{\gamma}^n$, in which σ_y, k, and n are the yield stress, consistency, and power-law index, respectively. The quality of this description of the steady shear experimental data is evident in Fig. 2(a).

To describe the recoverable behavior, we define the recoverable modulus $G_{\rm rec}$, in terms of the stress and recoverable strain determined via oscillatory shear [13,21],

$$G_{\rm rec} = \frac{\sigma_0}{\gamma_{\rm rec,0}} \tag{5}$$

where σ_0 is the amplitude of the stress and $\gamma_{\rm rec,0}$ is the amplitude of the recoverable strain. The recoverable modulus as a function of the stress amplitude determined from our experiments for each polymer concentration is shown in Fig. 2(b). Three clear regions of behavior can be identified: a small stress region where $G_{\rm rec}$ is independent of the applied stress; an intermediate stress region, where $G_{\rm rec}$ decreases with increasing stress; and a large stress region where $G_{\rm rec}$ is an increasing function of stress. We can describe these experimental data by combining a power-law term that defines the softening behavior and a finite extensibility term that accounts for the increasing stiffness at large stress,

$$G_{\rm rec}(\sigma) = G_\infty + \frac{G_0 - G_\infty}{1 + \left(\frac{|\sigma(t)|}{\sigma_c}\right)^{n_x}} + \frac{|\sigma(t)|}{\gamma_{r,0,\max}}. \tag{6}$$

Here, G_0 is the modulus at small deformations, G_∞, σ_c, and n_x are used to define the power-law softening, and $\gamma_{r,0,\max}$ is the maximum recoverable strain the material can acquire. The first two terms of this expression have the form of the Cross model used to describe the shear thinning viscosity observed in many viscoelastic fluids [67], with our subject being the modulus and our stress playing the role of Cross's shear rate. The final term of Eq. (6) is a direct result of finite extensibility, and means that once the material has reached its maximal recoverable strain, any additional strain will be acquired unrecoverably. Since the recoverable strain reaches a maximum but the applied stress can continue to increase, the recoverable modulus increases with a slope of unity on a log-log scale. We show in Fig. 2(b) that Eq. (6) provides an excellent description of the experimental data.

Having established the recoverable and unrecoverable behaviors separately, we now combine them to complete the construction of the model using the Kamani-Donley-Rogers (KDR) framework. In the KDR description of yield stress fluid rheology, rapid acquisition of recoverable strain enhances plasticity [18]. A recent extension of the KDR model includes a parameter that accounts for brittle and ductile yielding by adjusting the amount by which recoverable strain acquisition enhances plasticity [19]. We use the same concept here. A 1D version of our complete model is

$$\sigma + \left(\frac{\eta_f(\dot{\gamma}_r,\dot{\gamma}_u)+\eta_s}{G_{\rm rec}(\sigma)}\right)\dot{\sigma} = \eta_f(\dot{\gamma}_r,\dot{\gamma}_u)\left(\dot{\gamma} + \frac{\eta_s}{G_{\rm rec}(\sigma)}\ddot{\gamma}\right), \tag{7}$$

where σ and $\dot{\sigma}$ are the stress and its temporal derivative, and $\dot{\gamma}$ and $\ddot{\gamma}$ are the first and second time derivatives of the strain. $G_{\rm rec}(\sigma)$ is the stress-dependent modulus shown in Eq. (6), and η_s is the viscosity associated with recoverable dissipation of energy, which is determined as G''/ω at $\omega = 1$ rad/s in the linear regime. The rate dependent flow viscosity is given by η_f as

$$\eta_f(\dot{\gamma}_r,\dot{\gamma}_u) = \frac{\sigma_y}{\left|\frac{\dot{\gamma}_r}{Bt} + \dot{\gamma}_u\right|} + k\left|\frac{\dot{\gamma}_r}{Bt} + \dot{\gamma}_u\right|^{n-1} \tag{8}$$

where σ_y, k, and n are the yield stress, consistency, and power-law exponent obtained from the Herschel-Bulkley

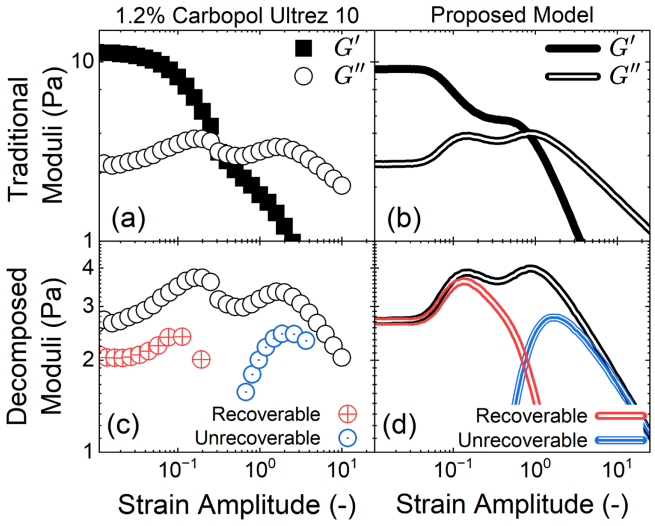


FIG. 3. (a),(c) Experimental data and (b),(d) proposed model predictions for Carbopol Ultrez 10 in propylene glycol at 1.2 wt %. (a),(b) Total storage modulus (filled squares/solid line) and total loss modulus (open circles/open line) as a function of strain amplitude. (c),(d) The total loss modulus (open circles/line), and the recoverable (red circles filled with plus/open line) and unrecoverable (blue circles filled with dot/open double line) contributions.

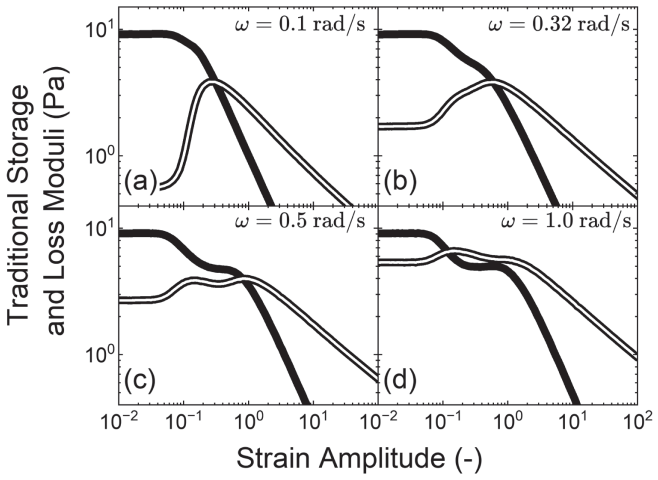


FIG. 4. Proposed model predictions for total storage and loss modulus at different angular frequencies, showing the frequency dependence of the two-step phenomena. (a) 0.1, (b) 0.32, (c) 0.75, and (d) 1.0 rad/s.

description of the steady state behavior shown in Fig. 2. $\dot{\gamma}_r$ and $\dot{\gamma}_u$ are the recoverable and unrecoverable strain rates, and Bt is the brittility factor. Determination of the Herschel-Bulkley parameters comes from fittings to the steady-shear flow curve shown in Fig. 2(a). A full tensorial representation is shown in the Supplemental Material [63].

We show in Fig. 3 excellent quantitative agreement between the experimental data and the model predictions in terms of the traditional moduli and their components. Both the experimental and model storage modulus show a linear viscoelastic regime at small amplitudes, and two separate regions of decrease. The first decrease is due to the softening of the elastic modulus, which results in less energy being stored for a given strain amplitude. The second decrease has a different physical cause and is due instead to the inverse square dependence of the storage modulus on the total strain amplitude shown in Eq. (1). When yielding, the biggest cause of change in the storage modulus is not actually energy storage but rather the significant increase in the unrecoverable strain, which acts to increase the denominator. The storage modulus therefore drops because the square of the total strain increases faster than the energy being stored increases.

The loss modulus obtained from our experiments and our model also show linear viscoelastic plateaus, and then the same two distinct overshoots observed previously [38]. The underlying physics which led to the model construction are also accurately accounted for, as observed in the components of the loss modulus. The first overshoot is clearly associated with an increase in the recoverable component of the loss modulus, while the second is associated with the unrecoverable component. This two-step behavior is therefore reflective of an elastic softening at smaller amplitudes followed by yielding at larger amplitudes. The underlying physics that causes the two overshoots is distinctly different.

A generic description of the softening observed in these systems is related to the attractive nature of the constituents. At small stresses, the recoverable modulus is independent of stress, indicating little separation of the constituents, and an attractive potential which is effectively constant. As the stress increases, the constituents begin to experience larger separations, leading to a decrease in the attraction and therefore a smaller recoverable modulus. More strain is therefore acquired recoverably. As the stress continues to increase, the separation becomes large enough to exceed the attractive part of the potential and additional deformation is acquired unrecoverably.

An interesting test of our model's predictive capabilities is to see what it predicts as we vary the angular frequency. The predictions of the loss modulus are shown in Fig. 4. Previous works have shown a frequency dependence to the size and appearance of the two-overshoot response in the dynamic loss moduli [33,38]. Our model captures the same phenomena. The magnitude and separation of the two overshoots depend on the angular frequency. At low frequencies, our model shows a single overshoot, while two overshoots are clearly observed at higher frequencies. The reason for this behavior is that the two underlying phenomena occur at similar strain amplitudes at lower frequencies, leading to a total loss modulus that appears as a single smooth curve from which the two effects cannot be resolved. The peak associated with yielding occurs at smaller strain amplitudes at lower frequencies because the model has a rate-dependent yield strain [18]. In the

limit of zero frequency this yield strain becomes the quasistatic value [19] of $\gamma_{y,\omega\to 0} = G/(Bt\sigma_y)$. By contrast, the peak associated with the elastic softening does not change its position as the frequency is decreased because it is a purely strain-related effect. The elastic term is modeled after a Kelvin-Voigt solid, where the total stress is a sum of an elastic and a viscous stress. For a given strain, a lower frequency means the strain rate is smaller, leading to a diminishment of the viscous stress. At lower frequencies, therefore, the stress is dominated by the elastic stress. The critical strain at which this effect occurs is σ_0/G_{rec}, which is independent of the angular frequency. Data for the various polymer concentrations, shown in the Supplemental Material [63], support these conclusions.

This Letter sheds new light on the complex yielding behavior of soft materials, particularly in relation to the manifestation of multiple overshoots in the dynamic loss modulus during oscillatory amplitude sweeps. Through recovery rheology experiments and the construction of a rheo-physical model, we have shown that the dual overshoot phenomenon can reflect two distinct phenomena. In our Letter, the first overshoot reflects elastic softening, and is associated with the acquisition of recoverable strain, while the second overshoot is directly linked to the yielding process, indicative of a transition to unrecoverable acquisition of deformation. While the physics underlying other two-step yielding systems may differ from that elucidated here, it is clear that recovery rheology techniques will play an important role in resolving the underlying physics governing the behavior of complex materials. By isolating the effects of recoverable and unrecoverable deformation, we have been able to elucidate the complex interplay between nonlinear elastic behavior and the yielding transition in yield stress fluids exhibiting the two-overshoot phenomenon.

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Data availability—The data are available from the authors upon reasonable request.

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